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Advancing Our Toolbox for Quantitative Carbonate Diagenesis Studies on the Arabian Platform: Insights from LA-ICP-MS *In-Situ* U/Pb Chronometry and Clumped Isotope Geothermometry

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Abstract

Diagenesis plays a key role in shaping pore space and the petrophysical properties of carbonate successions. Traditional petrographic analyses often provide broad relative timelines with limited quantitative constraints. This study explores the potential of advanced geochemical techniques—namely *in-situ* LA-ICP-MS U/Pb dating and clumped isotope geothermometry—to develop a thermo-chronometric approach to diagenetic research, with a specific focus on carbonate successions in Saudi Arabia.

Five distinct stratigraphic units, spanning from the Permian to the Eocene, were sampled from outcrop and subsurface locations across Saudi Arabia. In-situ LA-ICP-MS U/Pb chronometry provided absolute age dating for various diagenetic phases, including replacive dolomites, dolomite cements, calcite cements, and micritic matrix. Clumped isotope ($\Delta47$) analysis was conducted to determine precipitation temperatures, especially in phases where traditional fluid inclusion techniques are not applicable—such as micritic matrix and finely crystalline dolomites. These geochemical results were integrated with stable isotope (C/O) analysis, petrographic observations, and basin burial, thermal, and fluid history to reconstruct the diagenetic/geodynamic evolution of the studied basin.

Across the five stratigraphic intervals, comprising 45 samples, U/Pb analysis achieved a success rate of up to 90%, providing robust evidence for a complex diagenetic history within each unit. Absolute U/Pb ages range from 265 ± 8.1 Ma to 193 ± 15.7 Ma in the Khuff Formation (Permian), 230 ± 22 Ma to 135 ± 20 Ma in the Jilh Formation (Late Triassic), 98.6 ± 4.1 Ma to 26 ± 0.66 Ma in the Wasia Formation (Albian-Cenomanian) and 59.6 ± 13 Ma to 1 Ma in the Rus Formation (Ypresian). The diagenetic phases are generally younger than their respective stratigraphic ages, consistent with their diagenetic nature, yet still indicating a relatively early diagenetic onset. In the Khuff and the Rus formations, U/Pb ages suggest syn-depositional timing, offering insights into early diagenetic processes. Across all units, U/Pb ages exhibit high consistency with petrographically determined paragenetic sequences, allowing calibration of the entire paragenesis into an absolute temporal framework. Coupling these ages with clumped isotope data enables reconstruction of paleothermal regimes, locally highlighting low-temperature recrystallization of micritic and dolomitized matrix, but also revealing hydrothermal activity and fault-related fluid migration.

The dataset collected in this study, through combined *in-situ* U/Pb chronometry and clumped isotope analysis, underscore the complementary value of these techniques in unraveling complex carbonate diagenetic histories, particularly dolomitization. The findings have significant implications for quantifying pore space evolution using absolute thermo-chronological data, ultimately aiding in reservoir modeling and prediction of rock types at the platform scale.

Introduction

In both outcrops and subsurface, diagenesis has a major influence on the generation, preservation and destruction of the pore volume within sedimentary rocks and particularly in carbonate strata. Diagenetic processes such as dissolution, cementation, fracturing, *etc*., are also directly related to basin-scale fluid migrations (water or hydrocarbons), providing key insights on basin evolution through time. For decades, diagenetic studies have been mainly based on standard thin-section petrography (usually including carbonate staining and cathodoluminescence) that provide a primary understanding of the various phases along with their broad cross-cutting relationships and broad assumption on parent fluids. Geochemistry, including stable C/O isotopes, $^{86}Sr/^{87}Sr$ and rare earth element concentrations are classically used to refine the origin and timing of calcite and dolomite cements, providing important information on parent fluids (e.g. marine, meteoric, burial, *etc*). Ultimately, fluid inclusion studies are mainly used to reconstruct crystallization temperature and salinity of the parent fluids, where cements are large enough to host inclusions. All these techniques are of course extremely valuable but only provide broad relative timings for the various studied phases and most of the time rely on ‘educated guesses’ to determine timing of precipitation (i.e, by comparing fluid inclusion temperatures with burial history curves, *etc*).these techniques also rely on relatively large cement phases that can easily be sampled for isotopes and are more likely to have fluid inclusions, leaving fine-crystalline material such as micritic matrix or early cements widely overlooked.

Over the last decade, the fast development of cutting-edge technologies such as LA-ICP-MS in-situ U/Pb chronometry on calcite and dolomite cements and clumped isotopes (Δ47) drastically improved the understanding of the timing and origin of basin-scale diagenetic processes. Pb chronometry for example enabled the dating of key stratigraphic surfaces, fault generation and reactivation, meteoric fluid recharges, dolomitization events *etc*, with potentially limited error margins. Clumped isotopes on the other hand, have proven very useful to provide precipitation temperature and direct parent fluid composition, in addition to fluid inclusions, or in samples where fluid inclusions were not present (i.e., fine crystalline micritic matrix, replacive dolomite, *etc*).

In this context, the aim of this study is to explore the potential of both *in-situ* U/Pb chronometry and clumped isotopes on a wide variety of carbonate units in Saudi Arabia to improve the overall understanding of the diagenetic processes that affected the subsurface and outcrop strata of the Arabian Platform.

Geological background and studied material

In this study, five distinct stratigraphic units have been investigated spanning from Permian to Eocene, in both subsurface and outcrops in Saudi Arabia, covering ~270Ma of geological time. A summary of the depositional setting of each unit is provided below (in stratigraphic order).

- The oldest analyzed unit corresponds to the Lopingian ***Khuff Formation*** (~265Ma). Studied deposits are pervasively dolomitized mudstones and coated-grain rich grainstones deposited in a shallow marine inner ramp setting. Five samples were analyzed for *in-situ* U/Pb chronometry, targeting dolomite cements and replacive dolomite fabrics.
- During the late Triassic, the uppermost part of the ***Jilh Formation*** (~235Ma) was deposited in a peritidal to shallow marine, low-energy inner ramp environments where facies mainly comprise

dolomitized mudstones and peloidal packstones. For this interval, 10 samples were selected for U/Pb chronometry, along with 4 clumped isotopes, with the main aims to characterize dolomitization processes through the burial history.

- The intertidal facies of the early Valanginian ***Buwaib Formation*** (~140Ma) are characterized by pervasively dolomitized mudstones and microbial bindstones, thought to be the result of an early dolomitization process (Deville de Periere et al., 2024). U/Pb chronometry (3 samples) and clumped isotopes (4 samples) were conducted to assess the timing of the dolomitization and potential recrystallization during burial.
- In the Albian/Cenomanian strata of the ***Wasia Formation*** (~97Ma), deposits alternate between basinal facies rich in planktonic foraminifera and shallow marine inner ramp facies with benthic foraminifera and green algae. The diagenetic history of this section involves several major unconformities related to regional uplifts and in-situ U/Pb chronometry (15 samples) was used to understand the diagenetic impact of each unconformity.
- Ultimately, the Ypresian ***Rus Formation*** (~53Ma) was studied in the outcrops of the Eastern Province to reconstruct the diagenetic origin of large calcite and quartz filled cavities and of the surrounding dolomitized matrix. These peritidal deposits were examined through the analyses of five U/Pb samples and three Δ47 samples.

Advanced analytical tools for carbonate diagenesis

LA-ICP-MS in-situ U/Pb chronometry

In-situ U-Pb dating of carbonates by LA-ICP-MS methodology has a proven potential to drastically improve diagenetic studies by providing absolute age dating of the various carbonate phases (micritic matrix, calcite and dolomite cements, etc). These data can then be related to basin-scale fluid flow models. U/Pb chronometry is, however, not yet widely used on the Arabian Platform, with only limited published studies in the Mesozoic (e.g., Gasparrini et al., 2023; Mansurbeg et al., 2024) and Cenozoic (Markowska et al., 2025). Hence, the present study is the first of its kind on the Arabian Platform to include various stratigraphic units and a wide range of diagenetic phases.

U-Pb dating was conducted at CEREGE, France. A total of 43 samples has been used to analyze the various stratigraphic unit described above. Thin sections (60 µm) were screened to identify zones with sufficient U and Pb signals and variable U/Pb ratios. Analyses used a NWR193 laser (193 nm) coupled to a Thermo Element XR SF-ICP-MS, following Horstwood et al. (2016). Ablation was performed with a 150 µm spot, 15 Hz repetition rate, and 0.9–1.4 J/cm^2 fluence, using helium and argon as carrier gases. The ICP-MS measured isotopes 206Pb, 207Pb, 208Pb, 232Th, and 238U, with signal integration times of ~0.04 s per peak. Data were processed in Iolite 3.65, using on-peak gas blank subtraction, log-ratio calculation, and drift correction with NIST-614 and carbonate standards (RA138 for 206Pb/238U, AUG-B6 as secondary). Ages were calculated with IsoplotR using Tera-Wasserburg plots and model-1 regression, reported at 95% confidence. Common Pb correction was not applied. Validation was achieved using AUG-B6, yielding 42.8 ± 2.6 Ma (2SD, n = 12), consistent with literature values. Multiple ablation tracks were placed across micritic, sparitic, and diagenetic features to test isotopic homogeneity and assess timing of recrystallization.

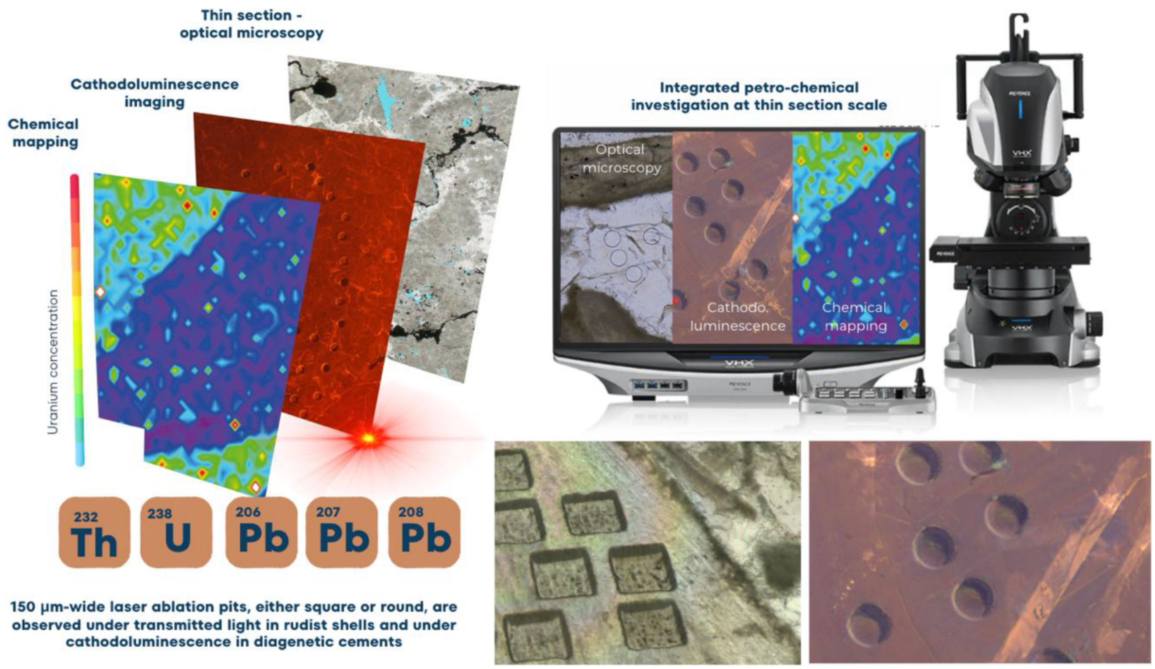

Figure 1—Integrated petro-chemical investigation of carbonate thin sections using transmitted light, cathodoluminescence imaging and LA-ICP-MS chemical mapping prior to U/Pb dating. The square and round ablation pits are used for in situ U/Pb carbonate geochronology, here applied in both primary biogenic phases (e.g. rudist shells) or latter diagenetic phases/cements.

Signal intensity was monitored in real-time to avoid inclusions and fractures, which were identified petrographically and excluded from final age interpretation. Each analytical session included replicate analyses of reference materials interspersed with unknowns to ensure reproducibility. Laser and ICP-MS performance were evaluated daily through calibration routines and sensitivity monitoring, maintaining within-run and between-run precision. Background levels of U and Pb were minimal, and downhole fractionation was systematically evaluated but found negligible under the applied conditions. Resulting ages are interpreted within their petrographic and paragenetic context.

Clumped isotope geothermometry

Classically, the temperature of precipitation of carbonate cements is measured using fluid inclusions and the initial parent fluid composition is back-calculated from the measured temperature and oxygen isotope values measured on a given cement. Clumped isotopes (Δ47) offer a very powerful and time/cost-efficient alternative to fluid inclusions, notably in phases where such inclusions are very challenging to find, or are not present (e.g. micritic matrix, fine-crystalline dolomitized facies). Therefore, Δ47 analyzes enable the characterization of more phases than fluid inclusions and Δ47-derived temperatures of these carbonates can be compared with burial history curves. Hence, Δ47 appears to be a key tool for modern diagenetic studies. On the Arabian Platform, clumped isotopes are more widely available in the public domain compared to U/Pb data (e.g., Sena et al., 2014; Vahrenkamp et al., 2014; Barata et al., 2015; Gasparrini et al., 2023; Herlambang & John, 2023; Bergmann et al., 2025; amongst others) and are of critical importance to reconstruct basin-scale paleothermal fluid flows.

Carbonate samples including micritic matrix, dolomitized matrix and calcite cements, were micro-drilled using a 0.5 mm carbide bit to avoid mixing. The various analyzed calcite cements were initially identified using petrography, including carbonate staining and cathodoluminescence. Between 2–10 mg of powder was collected for each sample depending on phase size. Clumped isotope analyses were performed at Massachusetts Institute of Technology (MIT) on a Nu Perspective IRMS following Anderson et al. (2021). Carbonates (~400–600 µg) were reacted with phosphoric acid at 70°C for 25 minutes. The evolved CO_2 was purified cryogenically and with a Porapak Q trap. Sample and reference gases were measured alternately across m/z 44–49 using 3 acquisitions of 20 cycles each (30 minutes total). Analytical runs included ETH-1–4 standards and IAEA-C1/C2 for quality control, maintaining an approximate 1:1 ratio of unknowns to anchors. Cycles with Δ47 values >5 SD from the median were removed (0.2%), and entire analyses were excluded if >10 cycles were outliers, pressure <15 mbar, or >1% imbalance occurred. Data were processed with "D47crunch" using IUPAC 17O parameters and acid fractionation factors, projected to I-CDES with reference to InterCarb values. Final Δ47 values were calculated via pooled regression. IAEA-C1 was used to check internal consistency; long-term repeatability and dataset details are reported elsewhere. The dataset was subjected to rigorous internal calibration, with each batch incorporating randomized ETH anchor sequences to avoid bias. Particular care was taken in sample gas handling to mitigate memory effects and ensure full cryogenic recovery. Baseline corrections were not applied, as pressure stability and signal decay remained within acceptable thresholds throughout the campaign. Instrument performance was monitored via daily calibrations and periodic cross-checks with historical runs, ensuring data comparability across sessions.

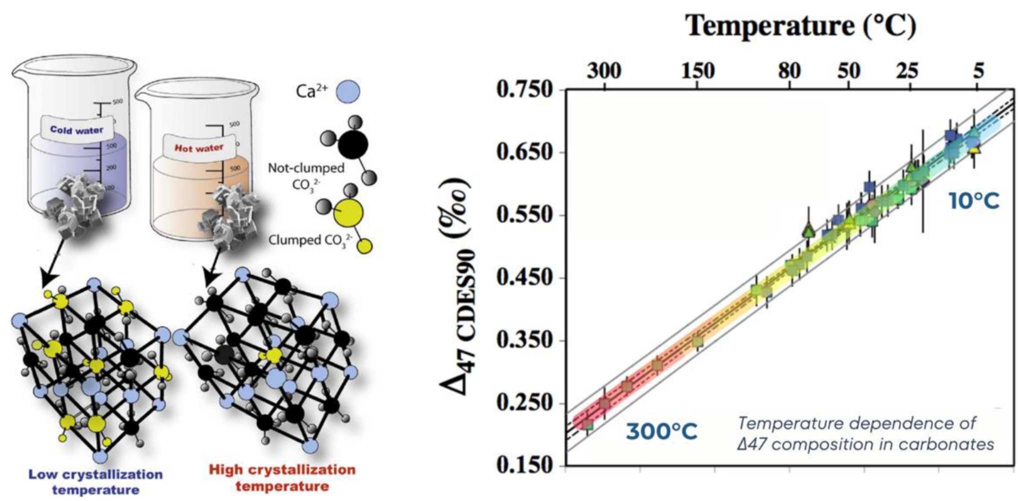

Figure 2—This figure illustrates the principle and application of clumped isotope thermometry in carbonates. On the left, carbonate crystals precipitating from cold water show a higher abundance of clumped isotopologues (e.g., ^{13}C-^{18}O bonds in CO_3^{2-}), while those from hot water show fewer clumps. On the right, the Δ_{47} (CDES90) values measured in carbonates are plotted against formation temperature, demonstrating a robust inverse relationship. Higher Δ_{47} values correspond to lower crystallization temperatures, enabling reconstruction of past temperatures based on carbonate composition.

Applications

Dolomitization and dolomite cements

A large amount of the studied samples has been used to provide a time constrain on dolomitization processes, notably within the Upper Permian Khuff Formation, late Triassic Jilh Formation and in the Ypresian Rus Formation. In both Khuff and Jilh formations, U/Pb chronometry suggests that the dolomitization events have a younger age than deposition. In the Khuff Formation for example, Deville de Periere et al. (2025a) indicate that the observed replacive dolomite and usually interpreted as an early diagenesis process has a broad early Triassic age (average 244.3 ± 7 Ma - Figure 3). These values, however, when compared to stable isotopes indicate a recrystallization event of the dolomite rather than a primary signal. In the Jilh Formation, Deville de Periere et al. (2025b) show at least two phases of dolomitization: rD1 with a Late Triassic age (208 ± 16 Ma, n = 7) and rD2 with a precipitation during the early Jurassic and an age of 177± 13 Ma (n=5). The values measured on the replacive dolomites of the Jilh Formation were critical as they enabled the precise dating of the various rock-types and associated pore system observed in the studied unit.

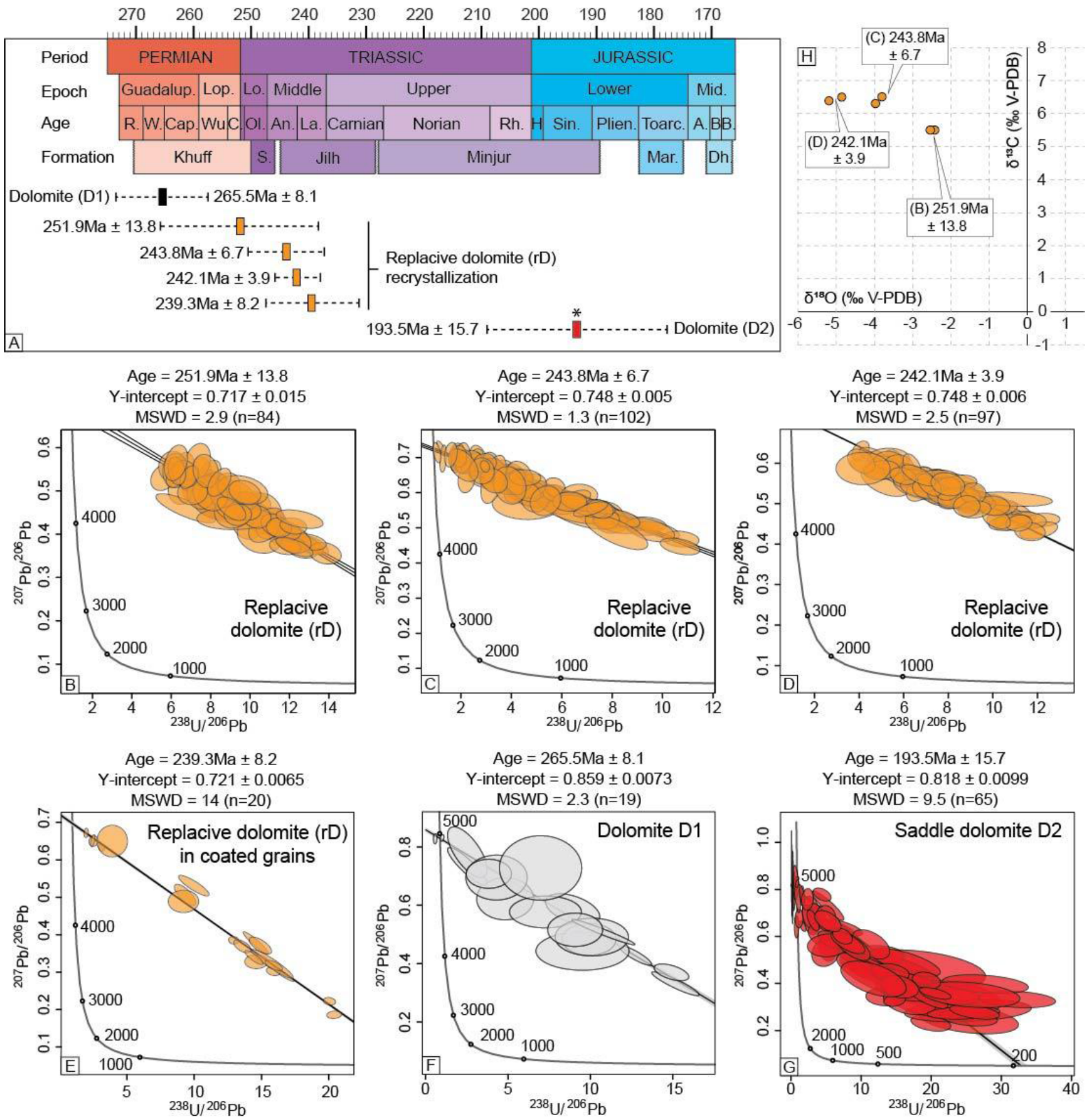

Figure 3—Example of U/Pb age dating on dolomites from the Khuff formation of Saudi Arabia (Deville de Periere et al., 2025). (A) Compiled results of the *in-situ* U/Pb age dating with Tera-Wasserburg Concordia diagrams and corresponding lower intercept ages for the various dolomite phases (B-G). Data point error ellipses indicate 2σ internal uncertainty of the isotope ratios on "n" analyses. MSWD—mean squared weighted deviates. Black star for D2 indicate sample with fluid inclusion analyses. (H) Oxygen/Carbon stable isotope cross plot showing the distribution of the replacive dolomite analyzed samples, along with their respective U/Pb ages.

In the outcrops of the Rus Formation, U/Pb data indicate an average crystallization age of 54.6 ± 7Ma, which is broadly consistent with the Ypresian deposition age of the Rus Formation. U/Pb ages, however, spread from 59.6 ± 13.1Ma to 52.6 ± 7.9Ma (n=3) and Δ47 data suggest a precipitation temperature of ~45°C likely confirming early dolomitization and recrystallization process under limited burial depth. Dolomite cements such as large saddle dolomite appeared more challenging to date than the replacive dolomites. In the dolomitized Buwaib Formation, the U/Pb chronometry provided unreliable ages, with very large error marking (usually more than 30Ma), due to the high abundance of common lead. Clumped isotopes, however, show a clear temperature gradient, in which temperature increases along with decreasing oxygen isotopes. In this case, dolomitized mudstones with $\delta^{18}O$ values between +1‰ and +2‰ V-PDB precipitated at temperature around 30°C, very close from depositional conditions, hence suggesting a syn-sedimentary dolomitization process. Dolomites with $\delta^{18}O$ values between -1‰ and +0.5‰ V-PDB and fairly similar petrographical characteristics, precipitated at temperature between 40°C and 60°C suggesting recrystallization under limited burial.

In both Khuff and Jilh formations, hydrothermal dolomites were dated in a limited number of samples and with locally rather high uncertainty (e.g. ± 15.7 Ma in the Khuff Formation, representing ~8% of the measured age – Deville de Periere et al., 2025a). These data, however appeared critical as rare oil inclusions were present in these crystals, providing an insight on hydrocarbon migration timings. Furthermore, these hydrothermal phases are thought to be associated with fluid migration along faults systems. Hence, the U/Pb chronometry data from such burial cements can support basin-scale modeling of fluid migrations.

Linking calcite cements to the regional tectonostratigraphic context and paleoclimates

The post deposition history of Wasia Formation (Albian/Turonian) is marked by the influence of several regional unconformities associated with dissolution and cementation events. Using petrography and U/Pb chronometry, Deville de Periere & Mangenot (2025) show that at least five phases of calcite cements precipitated within the studied interval. Of major importance are calcites associated with the Pre-Aruma Unconformity dated from the Turonian and associated with exposures and fresh water influxes on shallow marine areas of the Arabian Platform (Deville de Periere et al., 2017). U/Pb chronometry on the calcites associated with this unconformity preserved an average age of 92.8Ma ± 3.6 and 87.7Ma ± 1.9, indicating at least two phases of fluid recharge during the Turonian and Campanian, possibly related to regional exposures. Calcite cements in large vugs and fractures indicated Ypresian (average 54.8Ma ±3.3) and Oligocene (25.4Ma ± 1.5) ages, suggesting calcite cementation associated with Alpine 1 and Alpine 2 uplift and fracture creation. In this section, it is also interesting to note that various calcite cements had similar petrographical characteristics but were clearly of a different age according to U/Pb data, highlighting the importance of using U/Pb chronometry along with standard observations.

Combining U/Pb age dating and clumped isotopes can also be very interesting to reconstruct paleoclimatic conditions on the Arabian Platform. For example, in the outcrops of the Rus Formation (Ypresian), abundant calcite concretions have been described, with calcite crystals reaching 2cm. These calcites preserve an average U/Pb age of 2.5 ± 3.3Ma (n=3), suggesting precipitation during the late Miocene to Pleistocene. Stable isotopes conducted on these calcites indicate very negative values with an average $\delta^{18}O$ of -11.5‰ V-PDB and average $\delta^{13}C$ of -3.2‰, initially suggesting a precipitation from hot fluids likely under burial conditions. The studied section, however, was very likely already cropping out or under very limited burial during the last 3Ma. Clumped isotopes on the calcite cements provided precipitation temperatures of ~45°C and a water composition of -5‰ to -7‰ V-SMOW suggesting meteoric parent fluids, likely brought by monsoon winds during humid phases over the last 3Ma. These results are perfectly in line with those published by Markowska et al., (2025) on speleothems of Saudi Arabia.

Challenges and pitfalls

LA-ICP-MS U/Pb carbonate dating and clumped isotope thermometry both offer powerful tools for reconstructing diagenetic and paleotemperature histories, but each method presents significant challenges. U/Pb dating of carbonates is complicated by heterogeneous uranium and lead distribution, the presence of common (non-radiogenic) lead, and the lack of matrix-matched standards for non-calcite matrix, all of which can compromise age accuracy. For example, in this study, hydrothermal saddle dolomites were usually challenging to properly date and U/Pb data provided unreliable results locally. Diagenetic alteration and misplacement of laser ablation pits in mixed phases further increases uncertainty. Clumped isotope thermometry (Δ47) is highly sensitive to post-depositional thermal reordering, which can reset isotopic signatures and obscure original formation temperatures in reservoirs subjected to intense thermal regimes (starting at vitrinite reflectance equivalence = 1.8%). Analytical precision is critical, and inter-laboratory discrepancies in calibration and correction protocols can limit comparability. Together, these pitfalls necessitate careful petrographic screening, cross-validation, and methodological rigor to ensure reliable data interpretations.

Conclusions

This study highlights the transformative potential of integrating LA-ICP-MS U/Pb carbonate chronometry with clumped isotope thermometry to reconstruct the diagenetic and thermal evolution of carbonate successions on the Arabian Platform. By applying these techniques across a wide stratigraphic and spatial range—from Permian to Eocene units in both subsurface and outcrop contexts—this study demonstrates their capacity to yield absolute age constraints and precipitation temperatures for diagenetic phases that are often overlooked using conventional methods. The high-resolution U/Pb age data align closely with petrographic parageneses, refining the timing of multiple dolomitization, calcite cementation, fault-related fluid migration and even hydrocarbon migrations. Clumped isotope data further complement this framework by offering temperature estimates where fluid inclusions are absent or ambiguous, enabling a more comprehensive reconstruction of past fluid regimes. The resulting dataset provides critical insights for rock type prediction, basin modeling, and understanding platform-scale fluid dynamics in complex carbonate systems.

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References

Anderson, N.T., Kelson, J.R., Kele, S., Daëron, M., Bonifacie, M., Horita, J., Mackey, T.J., John, C.M., Kluge, T., Petschnig, P., Jost, A.B., Huntington, K.W., 2021. A unified clumped isotope thermometer calibration (0.5-1,100 °C) using carbonate-based standardization. *Geophysical Research Letters* **48**, e2020GL092069.

Barata, J., Vahrenkamp, V., Van Laer, P.J., Swart, P., Murray, S., 2015. A regional analysis of clumped isotope geochemistry to define the timing of creation of micro-porosity in a Lower Cretaceous Giant Reservoir. SPE-177922-MS.

Bergmann, C., Osburn, M.R., Anderson, N.T., Hayhow, C., Wilcots, J., Cantine, M.D., Fisher, W.W., Boniface, M., 2025. Origins and alteration of Ediacarian carbonates recording the Shuram Excursion in Oman. *Geochemistry, Geophysics, Geosystems* 26. e2025GC012161.

Deville de Periere, M., Mangenot, X., 2025. Tectonostratigraphy of the Arabian Platform: Insights from U/Pb chronometry in calcite cements and microporous micrite. 86th EAGE Conference & Exhibition, Toulouse, France.

Deville de Periere, M., Mangenot, X., Dyja-Person, V., Aldhubaib, A., 2025a. Sedimentology and diagenetic evolution of the Upper Permian Khuff Formation in Saudi Arabia. New perspectives from biostratigraphy, U/Pb geochronology and fluid inclusions. *Marine and Petroleum Geology* **179**, 107462.

Deville de Periere, M., Mangenot, X., Dyja-Person, V., Muslem, A., 2025b. Diagenetic control on rock-type heterogeneity in dolomitized Upper Triassic Jilh Formation (Saudi Arabia) revealed by in-situ LA-ICP-MS U/Pb dating. SPE-227412-MS.

Deville de Periere, M., Durlet, C., Vennin, E., Caline, B., Boichard, R., Meyer, A., 2017. Influence of a major exposure surface on the development of microporous micritic limestones – Example of the Upper Mishrif Formation (Cenomanian) of the Middle East. *Sedimentary Geology* **353**, 96–113.

Gasparrini, M., Mourad, D., Mangenot, X., Boniface, M., Mourad, S., Nader, F.H., Gerdes, A., 2023. Dolomite recrystallization revealed by Δ47/U-Pb thermochronology in the Upper Jurassic Arab Formation, United Arab Emirates. *Geology* **51**, 471–475.

Herlambang, A., John, C., 2023. Fluid flow evaluation revealed by carbonate clumped isotope thermometry along the fractures in a complex salt dome setting: study case (Jebel Madar, Oman). *Marine and Petroleum Geology* **152**, 106220.

Horstwood, M.S.A., Košler, J., Gehrels, G., Jackson, S.E., McLean, N.M., Paton, C., Pearson, N.J., Sircombe, K., Sylvester, P., 2016. Community-derived standards for LA-ICP-MS U-(Th-)Pb geochronology – uncertainty propagation, age interpretation and data reporting. *Geostandards and Geoanalytical Research* **40**, 311–332.

Mansurbeg, H., Alsuwaidi, M., Mourad, D., Mourad, S., Tiepolo, M., Shahrokhi, S., Al-Aasm, I.S., Koyi, H., 2024. Disconformity-controlled hydrothermal dolomitization and cementation during basin evolution: Upper Triassic carbonates, UAE. *Geology* **52**, 486–491.

Markowska, M., Vonhof, H.B., Groucutt, H.S., Breeze, P.S., Drake, N., Stewart, M., Albert, R., Andrieux, E., Blinkhorn, J., Boivin, N., Budsky, A., Clark-Wilson, R., Fleitmann, D., Gerdes, A., Martin, A.N., Martinez-Garcia, A., Nicholson, S.L., Price, G.J., Scerri, E.M.L., Sholz, D., Vanwezer, N., Weber, M., Alsharekh, A.M., Al Omari, A.A., Al-Mufarreh, Y.S.A., Al-Jibreen, F., Alqahtani, M., Al-Shanti, M., Zalmout, I., Petraglia, M.D., Haug, G.H., 2025. Recurrent humid phases in Arabia over the past 8 million years. *Nature* **640**, 954–961.

Sena, C.M., John, C.M., Jourdan, A.L., Vandeginste, V., Manning, C., 2014. Dolomitization of Lower Cretaceous peritidal carbonates by modified seawater: constraints from clumped isotopic paleothermometry, elemental chemistry, and strontium isotopes. *Journal of Sedimetary Research* **84**, 552–566.

Vahrenkamp, V., Barata, J., Van LAer, P.J., Swart, P., Murray, S., 2014. Microrhombic calcite of a giant Barremian (Thamama B) reservoir onshore Abu Dhabi – Clumped isotopes analyses fix temperature, water composition and timing of burial diagenesis. SPE-172033-MS.